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Astrocyte-mediated higher-order control of synaptic plasticity

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Gustavo Menesse[1,2], Ana P. Millán[1] & Joaquín J. Torres[1]

The dynamics of higher-order topological signals are increasingly recognized as a key aspect of the activity of complex systems. A paradigmatic example are synaptic dynamics: synaptic efficacy changes over time driven by different mechanisms. Beyond traditional node-driven short-term plasticity, the role of astrocyte modulation through higher-order interactions, in the tripartite synapse, is increasingly recognized. However, the competition and interplay between node-driven and higher-order mechanisms remain poorly understood. Here, we introduce a higher-order model of the tripartite synapse, accounting for astrocyte-synapse-neuron interactions in short-term plasticity, such that astrocyte gliotransmission and pre-synaptic facilitation jointly modulate neurotransmitter release, generalizing earlier short-term plasticity models. We study these mechanisms in a minimal recurrent motif—a directed ring of three excitatory neurons—where one neuron receives external stimulation. Due to strong recurrence, the circuit is prone to self-sustained activity, often ignoring external input. By introducing higher-order interactions via astrocyte modulation, we show this robustly stabilizes circuit dynamics and expands the parameter space supporting stimulus-driven activity. Our findings highlight how astrocytes reshape effective connectivity through higher-order interactions—even in simple recurrent circuits.

Synaptic interactions between neurons form a dynamic, adaptive network that shapes brain dynamics and emergent cognitive phenomena. Such structure-function coupling is a defining feature of complex systems in which underlying connectivity shapes—and is shaped by—emergent behavior. Typically, the system dynamics are associated only with the nodes of the network, which in the case of neuronal networks stand for neurons. Recently, however, it has become evident that the edges between the nodes—the synapses—can also be associated with topological signals, i.e., dynamical variables which are not necessarily reduced to node dynamics[1]. Topological signals are ubiquitous in complex systems, and include molecular transportation fluxes in cells, synaptic signals in the brain, dynamic functional connectivity among brain regions[2–9], or vector fields such as the one representing currents in the ocean[10]. Each topological signal is associated with a dimension that emerges from the underlying topological substrate of simplicial and cell complexes[11] and reflects the type of interaction the signal represents: signals on nodes are 0-dimensional, those on edges are 1-dimensional, those on filled triangles or cycles (forming polygons) are 2-dimensional and so on. Simplicial complexes and topological signals provide a natural way to model and analyze complex, multi-scale interactions within a unified mathematical framework[11]. The emerging research field of higher-order topological dynamics[1], combining the study of non-linear dynamics with algebraic topology to model and study the dynamics of higher-order topological signals, has revealed the emergence of new dynamical states in such systems[12–17]. The impact of higher-order topological dynamics goes beyond the study of complex systems. Recently, for instance, it has led to a new general framework for AI algorithms that can take advantage of the novel dynamics that are uniquely possible for higher-order topological signals[18–22].

Synaptic signals are a paradigmatic example of higher-order topological signals. Synapses are typically modeled as the (1-dimensional) edges of neuronal networks and, notably, their efficacy changes dynamically over time both in short and long time-scales via synaptic plasticity mechanisms. Among these, short-term plasticity (STP) plays a crucial role in modulating the on-line transmission of the neural code by dynamically adjusting synaptic strength on timescales of milliseconds to seconds[23]. STP includes transient facilitation (STF) and depression (STD), which depend primarily on the recent activity of the presynaptic neuron and govern the immediate response of a synapse to successive stimuli[24–27]. This rapid, activity-dependent modulation acts as a dynamic filter that can emphasize or suppress specific temporal patterns of neural activity, effectively shaping

[1]Department of Electromagnetism and Matter Physics and Institute "Carlos I" of Theoretical and Computational Physics, University of Granada, Granada, Spain. [2]Departamento de Física, Facultad de Ciencias Exactas y Naturales, Universidad Nacional de Asunción, San Lorenzo, Paraguay.
e-mail: gmenesse@facen.una.py; jtorres@onsager.ugr.es

information flow in real time. A well-established model of STP introduced in Ref. 28 captures both facilitation and depression, and has been shown to generate non-equilibrium phases linked to dynamic memory and resonance phenomena near criticality[29–32]. In contrast to long-term plasticity, which involves lasting changes driven by correlated pre- and post-synaptic activity, STP provides a fast, flexible mechanism for regulating synaptic efficacy and supporting context-dependent computation in neural circuits.

Over the past two decades, accumulating evidence has shown that synaptic plasticity involves not only pair-wise, neuron-to-neuron communication, but also mechanisms that are intrinsically higher-order. A paradigmatic example is the tripartite synapse, in which astrocytes—non-neuronal glial cells and the most abundant cell type in the brain—play an active role in modulating synaptic transmission[33,34]. Unlike traditional neuron-centric models, where synaptic plasticity is viewed as a local, node-driven process, the tripartite model reveals a more distributed and collective form of regulation. Astrocytes can simultaneously monitor and modulate the activity of multiple neighboring synapses, often across different neurons, exerting control over synaptic efficacy and, consequently, over information flow in neural circuits[35,36].

Astrocytes regulate neuronal activity through a broad range of mechanisms (see[37] for a comprehensive review). These include direct modulation of synaptic plasticity via multiple signaling pathways[34,38], as well as indirect regulation through control of extracellular ion concentrations, such as potassium, which has been shown to play a key role in shaping neuronal bursting activity by modulating neuronal afterhyperpolarization[39]. These and other processes operate across multiple spatial and temporal scales and can strongly influence network stability and responsiveness. This glial modulation is not merely auxiliary but may constitute a core mechanism of synaptic computation, contributing to the temporal filtering, integration, and stabilization of synaptic signals[34,36]. The effects of astrocyte modulation compete with those of traditional node-driven STP, a fact that is typically overlooked both in STP and tripartite-synapse models. From a higher-order perspective, however, such effects arise naturally[1]. In this context, a neuronal network made of neurons and synapses is expanded by coupling the synapses through a new, higher-dimensional element, the astrocytes. Topologically, astrocytes are thus modeled as the 2-dimensional polygons of the higher-order network (or cell complex), in which nodes and synapses are respectively the 0-dimensional nodes and the 1-dimensional edges. And in a higher-order network modeling the dynamics of edge-signal (synaptic efficacies) naturally involves interactions through the nodes (neurons) they share, as well as through the polygons (the astrocytes) they conform[10,40].

Another example of higher-order modulation is the axo-axonic synapse, where a neuron directly targets the axon of another neuron to modulate the efficacy of its downstream synaptic connections—typically exerting an inhibitory effect[41,42]. Recent work has framed axo-axonic interactions as a form of triadic control over edge dynamics, leading to emergent spatiotemporal patterns in recurrent neural circuits[13,43]. Together, these mechanisms support the view that synaptic computation is not a purely dyadic phenomenon but a multi-agent, higher-order process, with astrocytes playing a particularly central role in the orchestration of dynamic and context-sensitive plasticity. In this context, a higher-order network perspective naturally arises, in which astrocytes mediate triadic interactions between synapses, giving rise to effective link-link couplings that cannot be reduced to independent pairwise synaptic effects.

From the neuroscience perspective, modeling of astrocyte signaling is a significant challenge due to their complex morphology and diverse dynamics. Astrocytes regulate synaptic plasticity by different mechanisms, including gliotransmission[44] and neurotransmitter uptake that clears the synaptic cleft[45]. In particular, gliotransmission is a process in neuron-astrocyte interaction by which astrocytes (and more generally glia cells) release gliotransmitters such as glutamate[46] and ATP[38] (key excitatory neurotransmitters) that interact directly with different receptors of both the pre- and post-synaptic neurons[44]. Among other effects[47–49] gliotransmission plays a role both in short-[50] and long-term[51] plasticity by modulating the probability of neurotransmitter release at the pre-synaptic neuron, either increasing or decreasing it[52,53]. Models of astrocyte regulation of different levels of detail have been proposed[45,54,55]. Many of these extend traditional short- and long-term synaptic dynamics to incorporate astrocyte influences, ranging from biophysically detailed compartmental models[56,57] to more simplified descriptions[58]. Overall, previous models have neglected the interplay between pre-synaptic and astrocyte mechanisms. Given than both of them act on the same neurotransmitters, exploring this competition remains an open question with potentially far-reaching implications.

In order to address this gap, in this study we propose a simple higher-order astrocyte-synapse-neuron short-term plasticity model (ASN-STP) that captures the competition between astrocyte and pre-synaptic mechanisms in short-term facilitation dynamics. Acknowledging the large variate of astrocytic regulatory mechanisms, we focus specifically on gliotransmission as a tractable and mechanistically transparent entry point to study higher-order astrocyte-synapse interactions. We introduce a general STP framework which reduces to previous models in the literature[28,50] for specific simplifications. By examining the dynamics of the tripartite synapse from a higher-order perspective[13], our model provides an unified framework to study the interplay between neuronal (dimension zero), synaptic (dimension one) and astrocyte (dimension two) dynamics, and how these shape emergent neuronal activity.

To illustrate the emergent dynamics of the system, we consider in particular the role of astrocyte regulation in a simple ring of neurons connected by excitatory synapses. Our findings illustrate that the presence of even a single tripartite synapse is sufficient to prevent runaway excitatory activity. This effect is strongest when higher-order interactions are present, such that a single astrocyte modulates several synapses coordinately, as it is the case in brain networks. Higher-order interactions further enhance the system's responsiveness to external stimuli when they act on internal synapses, as opposed to those responding directly external stimuli, suggesting that astrocyte modulation plays a more central role in deeper, integrative circuits, rather than sensory ones. The fact that tripartite synapse are more abundant in the hippocampus and cerebellum[59], present in more than 50% of all synapses in the hippocampus[60], which are regions with high internal recurrence[61,62] support the biological plausibility of our theoretical results. In short, our study indicate that higher-order astrocyte modulation is highly beneficial to control plasticity and neuronal activity of high-recurrent circuits.

Results

To study the low- and high-order effects of the tripartite synapse on synaptic and neuronal dynamics, we propose a neuron-synapse-astrocyte model (see "Methods" section). We consider a minimal recurrent circuit of excitatory neurons arranged in a ring, such that each neuron projects an excitatory (directed) synapse onto the following one (for simplicity, in the main part of this study we focus on this small recurrent network and in the final section we will explore the effect of increasing the circuit size). The network receives external input via a spike train arriving at one of the neurons, namely the read-in neuron, and activity is read from the last neuron in the ring, hereby refereed to as read-out neuron. As main control parameters of the model we consider the fraction α of neurotransmitters that remain available for the synapses—such that $(1-\alpha)$ is the fraction of neurotransmitters recruited by the astrocyte—and the frequency of incoming pulses at the read-in neuron.

This neuronal circuit already exhibits various behaviors in the absence of astrocyte interaction, driven by local STP (see Supplementary Fig. A), including synfire activity, bistable (switching on-off) activity controlled by external stimuli, and bursting activity. As shown in Fig. 1a, in this case the circuit is highly susceptible to entering a state of self-sustained activity (SSA) where the network is unresponsive to external stimuli—equivalent to a supercritical state[63]. This occurs as soon as synaptic currents are large enough so that each pre-synaptic spikes triggers a post-synaptic one. Consequently, the firing rate of the read-in neuron remains high for all values of α, and the SSA regime spans a large area of the parameter space (indicated by the black line in Fig. 1a). As we go on to show (see also

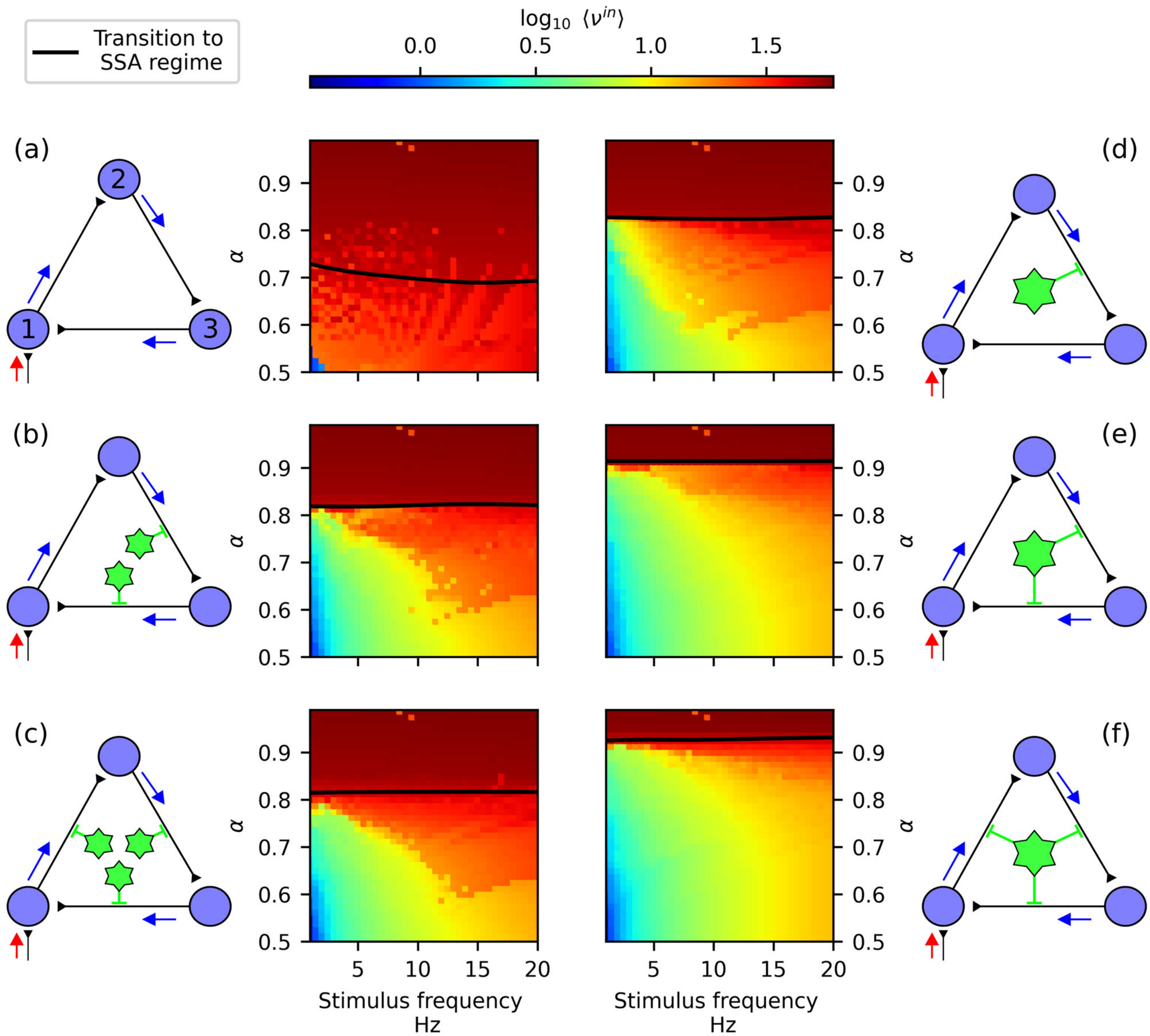

Fig. 1 | Astrocyte higher-order modulation extends the responsive region of the neuronal circuit. We show network activity through the average firing frequency of the read-in neuron $\langle \nu^{in} \rangle$ (heat-map diagrams) as a function of the fraction of neurotransmitters that remain in the synaptic cleft (α), and of the stimulus frequency. Each panel corresponds to a different astrocyte configuration as indicated by the corresponding graph diagram: **a** no astrocytes; **b**–**d** respectively one, two, and three astrocytes that interact with single synapses (low-order regulation); **e**, **f** one astrocyte that interacts with multiple synapses (higher-order regulation). In the graph diagrams blue circles stand for neurons, black lines for synapses (with the round end pointing towards the post-synaptic neuron), and green stars for astrocytes. Astrocyte-synapse interactions are indicated by green short lines. The read-in neuron that receives the external stimulus and which activity is shown in the heat-maps is indicated by a red arrow. In each diagram, the black line indicates the transition to the SSA regime. This line marks the critical value of α above which the synaptic currents emitted by neurons are always sufficient to trigger spikes in their postsynaptic partners, thereby sustaining activity within the circuit. For a detailed description of how the transition line is computed, see Supplementary Section 4. Network parameters were set to the default values listed in Table 1, and we set $\varepsilon = 0.01$, $\mathrm{Ca_{th}} = 0.04$, $\beta = 0.05$.

Supplementary Fig. B), astrocyte modulation disrupts the SSA regime and extends the region where the system is responsive to external stimuli.

In order to study the role of astrocyte modulation on preventing runaway activity, we focus here mainly on the case $\varepsilon < U_{SE}$. That implies, according to equation (11) (see “Methods” section), that astrocyte regulation in the tripartite plasticity leads only to a decrease in neurotransmitter release probability, i.e., a sort of anti-facilitation. In a later section, we also consider the effect of positive astrocyte facilitation ($\varepsilon > U_{SE}$). To characterize the system dynamics, we quantify the activity of the read-in (neuron 1, red arrow in Fig. 1) and read-out (neuron 3, see Fig. 1) neurons, denoted $\langle \nu^{in} \rangle$ and $\langle \nu^{out} \rangle$, respectively, to assess the potential functional benefits and side effects of negative astrocytic modulation.

Assuming the absence of gap-junction coupling between astrocytes ($D_{Ca} = 0$), we first compare different low- and higher-order modulation schemes. In low-order schemes, each astrocyte modulates only a single synapse, so that astrocytic dynamics effectively contribute only to first-order (link-level) interactions. In higher-order schemes, by contrast, each astrocyte modulates multiple synapses, thereby implementing effective link–link (second-order) interactions. This configuration is consistent with the organization observed in neuronal–astrocyte networks and naturally gives rise to higher-order interactions.

Finally, we consider astrocyte–astrocyte interactions mediated by gap junctions (see equation (17) in “Methods” section) and examine how these affect the system dynamics, and whether such diffusive coupling can recover

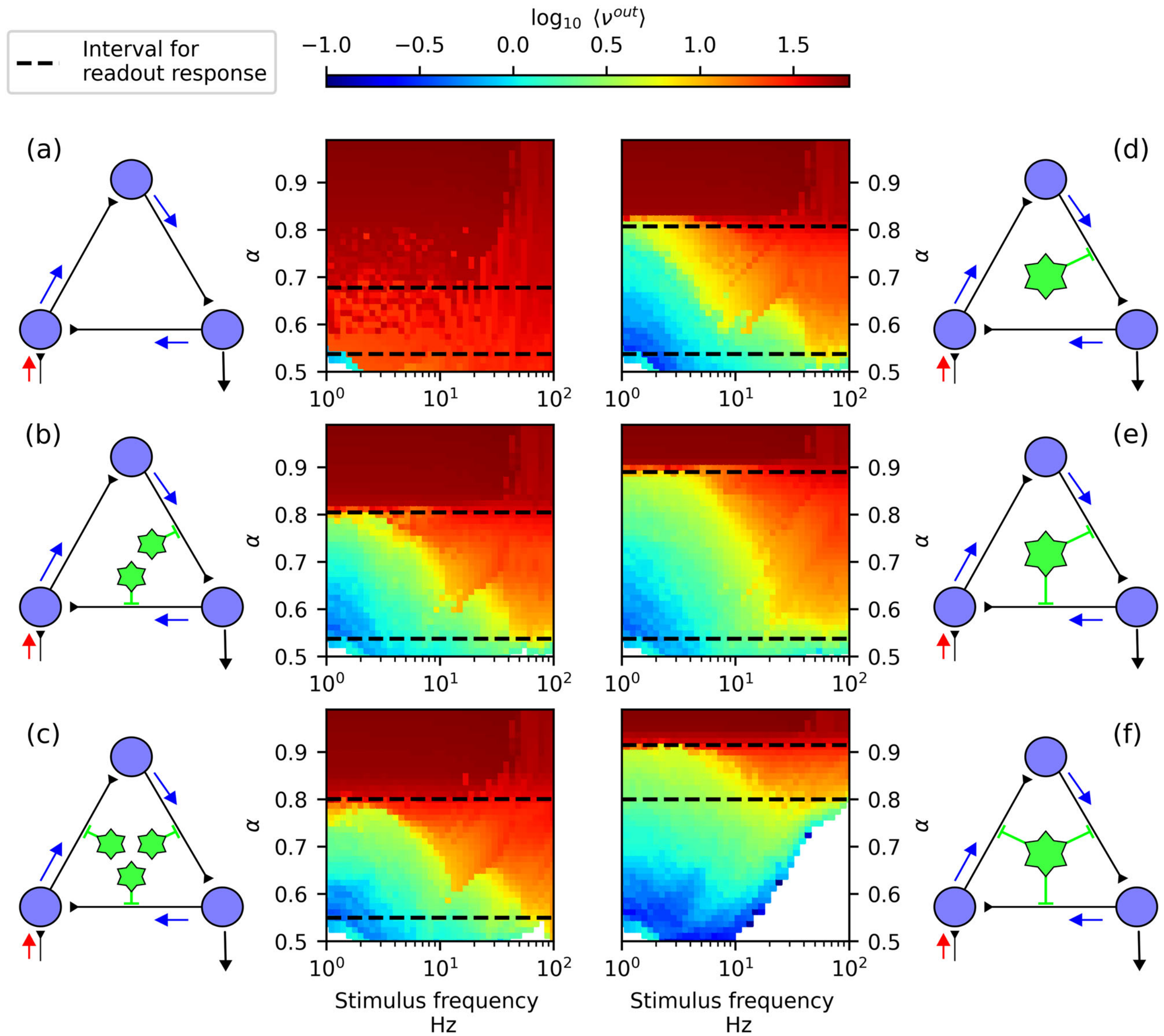

Fig. 2 | Astrocyte interaction scheme determines the circuit interval of monotonic positive response. We show network activity through the average firing frequency of the read-out neuron (indicated by a black arrow in the graph diagrams) $\langle\nu^{out}\rangle$. Each panel corresponds to a different astrocyte configuration as indicated by the corresponding graph diagram: **a** no astrocytes; **b–d** respectively one, two, and three astrocytes that interact with single synapses (low-order regulation); **e**, **f** one astrocyte that interacts with multiple synapses (higher-order regulation). The region between the black dashed lines indicates the range of α values for which the read-out neuron responds to the stimulus (below SSA transition) across the explored range of stimulation frequencies. Parameters: $\varepsilon = 0.01$, $Ca_{th} = 0.04$, $\beta = 0.05$. Other simulation parameters are listed in Table 1.

the effects of genuine higher-order astrocyte regulation in low-order schemes.

Higher-order modulation prevents runaway excitatory activity

We assess the regulatory effect of astrocyte modulation on activity propagation under low-order and higher-order schemes (see Fig. 1).

Our results show that the presence of even a single tripartite synapse is sufficient to regulate activity-propagation and significantly extend the area of the responsive phase in the parameter space, in particular in the control parameter α (panels b–f). Notably, this effect is greatest when astrocytes act in a higher-order scheme, that is, when a single astrocyte modulates multiple synapses (panels e, f), rather than astrocytes acting locally at the single-synapse level (panels b, c). In the higher-order scheme, astrocytes integrate the signals of different synapses. This not only leads to a faster $[Ca^{2+}]$ increase in the astrocyte and quicker gliotransmitter release, but this release also occurs simultaneously across all adjacent synapses, thus allowing for a more effective modulation of neuronal activity. These results highlight the role of higher-order interactions, as mediated by astrocytes in the tripartite synapse, in regulating excessive activity at the node and edge level, thus expanding the responsiveness of the system of external stimuli.

Higher-order tripartite plasticity improves read-out neuron response to stimuli

To explore possible side effects of astrocyte regulation, we consider in Fig. 2 the average firing-rate of the read-out neuron, namely $\langle\nu^{out}\rangle$, in the same modulation schemes as considered in the previous section (Fig. 1). In the absence of astrocyte regulation (panel a), $\langle\nu^{out}\rangle$ is high except for a small region for small α and small stimulus frequency (below 2Hz), for which the read-out neuron does not fire. In this regime a spike of the pre-synaptic neuron is not enough to cause a spike of the post-synaptic neuron, and STP does not effectively operate at such low frequencies due to the typical timescales of plasticity mechanisms[64]. Otherwise the average firing rate is high as for the pre-synaptic neuron in Fig. 1a, indicating low sensitivity of the circuit to the stimulation rate and its susceptibility to falling into the SSA regime. Similarly to the read-in neuron case, the presence of at least one tripartite synapse is sufficient to modulate activity, significantly increasing the read-out neuron's sensitivity to external stimuli.

To quantify the ability of the circuit to respond to external stimuli, we identify an interval in α for which the read-out neuron response monotonically increase (monotonic positive response) with respect to stimulus frequencies (1–100 Hz), and for which the circuit is not in the SSA regime. This is shown in Fig. 2 as the area between the two black dashed lines. A comparison between circuit activity in the safe regime with monotonic positive response vs SSA regime is presented in Supplementary Fig. C. The addition of astrocyte modulation, either in a low- or higher-order setting, expands the monotonic positive response interval except for the case where a single astrocyte regulates all three synapses (panel f), which we discuss in detail in the next paragraph. The configuration with the largest monotonic positive response interval (panel e) corresponds to higher-order regulation by a single astrocyte of the two internal synapses (synapses $2 \rightarrow 3$ and $3 \rightarrow 1$).

In the higher-order case in which the astrocyte regulates all three synapses (panel f), we observe that the read-out neuron becomes silent at high stimulation frequencies, and that the monotonic positive interval shifts toward higher α values while shrinking in extent.

Notably, as mentioned above, this phenomenon is not observed in the higher-order case in which the astrocyte does not regulate the first synapse (panel e). This is because the activation of the first synapse ($1 \rightarrow 2$) can be independent of the internal dynamics of the circuit, as it is directly driven by the external stimulus that induces firing in neuron 1. The repeated activation of synapse $1 \rightarrow 2$ maintains a high calcium concentration in the astrocyte (through the associated IP_3 currents, see equation (14) in "Methods"), causing the astrocyte to remain active for longer periods. This sustained activity reduces synaptic efficacy by lowering the release probability (equation (11) in "Methods"), thereby disrupting activity propagation before it reaches the read-out neuron.

A similar but significantly weaker effect occurs in the low-order modulation case (panel c), becoming noticeable only for very small values of $\alpha < 0.55$ and high stimulation frequencies (>50 Hz). In Supplementary Fig. D, we show that this effect arises in any scheme involving modulation of the first synapse, even when it is the only synapse being regulated. However, when the same astrocyte also interacts with the second synapse, the suppression of activity at high frequencies is considerably amplified. Although this frequency-selective response may be interesting from the perspective of selective signal transmission, usually studied in the context of excitatory-inhibitory circuits[65], in the following we mainly focus on the dynamical regime characterized by a monotonic positive response across stimulus frequencies.

In order to study the monotonic positive response regime in more detail, in Fig. 3 we show $\langle \nu^{\text{out}} \rangle$ again, but now using Poisson spike trains instead of periodic stimulation, and only for α values within the monotonic positive response interval identified in Fig. 2. The response is shown as a function of the stimulus frequency. In Supplementary Fig. E, we report the read-out response for different modulation schemes under Poisson stimulation. The results show the same qualitative behavior as in the periodic case, with a mitigation of the response suppression for low α and high-frequency observed in the low-order scheme due to the stochastic inter-spike intervals.

To further characterize the circuit response, we estimate the average dynamic range (see Supplementary Fig. F) of the read-out neuron for each modulatory scheme within the safe α interval (i.e., below the SAA transition). Using random rather than periodic stimulation allows us to obtain a smoother transfer function, which facilitates a more robust estimation of the dynamic range. The estimated values are reported as inset text in the corresponding panels of Fig. 3, together with the length of the read-out responsive α interval, $\Delta\alpha_r$. For additional details on the dynamic range estimation, see Section 6 of the Supplementary Information.

The presence of astrocyte modulation increases the average dynamic range within the safe region (responsive region below SAA transition). Low-order modulatory schemes lead to a more spread distribution of response firing rates for input frequencies below 10 Hz, indicating that small changes in α can produce abrupt increases in firing rate. This behavior makes the circuit less stable with respect to external stimulation, and this effect is more pronounced in the case of low-order modulation of the first synapse (see Fig. 3c).

In contrast, the higher-order scheme in which only internal synapses ($2 \rightarrow 3$ and $3 \rightarrow 1$) are modulated (see Fig. 3e) exhibits a high average dynamic range across the largest α interval. The higher-order case in which all three synapses are regulated (see Fig. 3f) shows the largest average dynamic range overall, although this occurs within a smaller α interval. In Supplementary Fig. G, we also report the dynamic range estimated for each value of α. This analysis makes explicit that astrocytic modulation extends the range of α values associated with a monotonic positive response (characterized by a large positive dynamic range), and provides a clear visualization of the relationship between the loss of monotonicity in the response and the SAA transition.

These results highlight a simple yet important point: for tripartite plasticity to be beneficial in a recurrent network such as the one studied here, modulation should only apply to internal synapses, i.e., synapses that are activated by the network's own activity. When astrocytes interact with synapses not regulated by the internal activity of the circuit, meaning, synapses that will be activated by external factors, then excessive modulation can occur, undermining the benefits of astrocyte involvement.

On whether higher-order modulation differs from low-order modulation

In our set-up, low- and higher-order regulation schemes differ in the number of synaptic inputs received by each astrocyte, which always equals one in the low-order schemes, and two or three in the higher-order ones. This leads to lower IP_3 production, lower calcium concentration, and thus reduced astrocyte modulation in the low-order schemes. This raises the question of whether the differences observed in Figs. 1 and 2 between low-order (panels b and c) and higher-order (panels e and f) regulatory schemes are simply due to a different scaling of the calcium levels, rather than an effect of higher-order regulation. To test this, in Fig. 4 we performed additional analyses for the low-order modulation scheme shown in panel (b) of Figs. 1 and 2, that is, the case where two different astrocytes regulate the internal synapses $2 \rightarrow 3$ and $3 \rightarrow 1$, respectively. In particular, we analyzed how the parameters β and Ca_{th} controlling the astrocytes' calcium dynamics affect the $\langle \nu^{out} \rangle$ diagrams. In particular, β controls the level of intracellular $[\text{Ca}^{2+}]$ elevation due to IP_3 production, while Ca_{th} sets the calcium threshold required for gliotransmitter release (equation (17) in Methods). Gliotransmission is facilitated both by increasing β and/or decreasing Ca_{th}. In Fig. 4 we systematically increase β and decrease Ca_{th}, starting from their reference values, thereby enhancing gliotransmission. We find that in either case the onset of the SSA regime (dark-red region) shifts towards higher α values, partially mimicking the behavior observed in the higher-order scheme of Fig. 2e. However, in neither case do we recover the length of α parameter interval for monotonic positive response $\Delta\alpha_r$ found under the higher-order regulation scheme. On the contrary, by increasing gliotransmission in the low-order case we find that it disrupts synaptic signal propagation prematurely, and the readout neuron becomes unresponsive to external stimuli (white region) already for smaller frequency inputs. Thus, the modulatory benefits seen in the higher-order setup cannot be replicated by simply amplifying astrocyte calcium currents in low-order schemes. This underscores the functional significance of astrocyte-mediated higher-order interactions in modulating synaptic activity at the circuit level.

Positive modulation by astrocytes

In the previous analyses, we assumed that gliotransmission exerts a depressive modulatory effect on synaptic transmission. More generally, however, gliotransmission can also have a facilitating effect. In the model, this is captured by the parameter ε: for $\varepsilon < U_{SE}$, gliotransmission decreases synaptic facilitation, whereas for $\varepsilon > U_{SE}$ it increases facilitation.

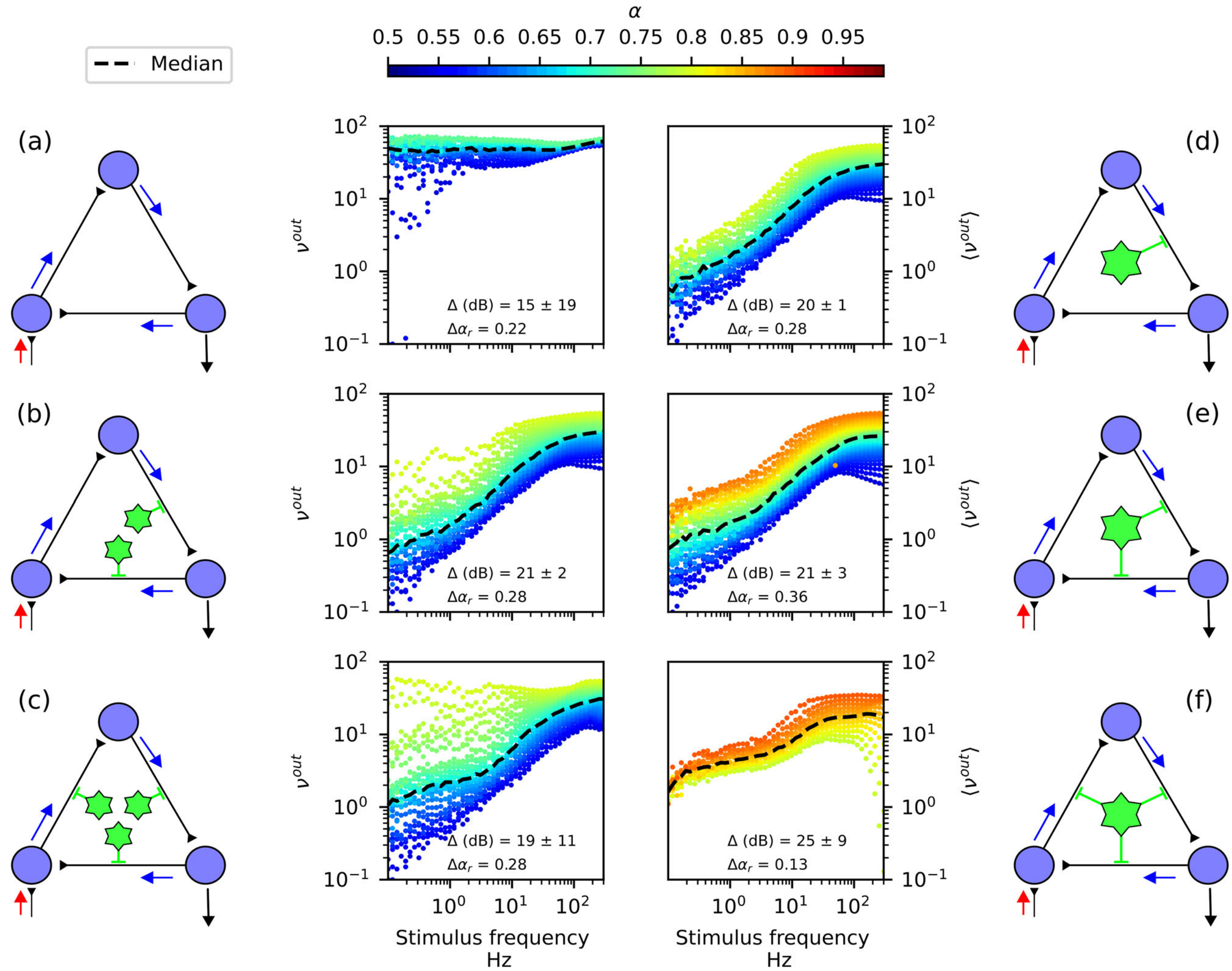

Fig. 3 | Neuronal circuit response improves under higher-order astrocyte modulation of internal synapses. Each panel corresponds to a different astrocyte configuration as indicated by the corresponding graph diagram: **a** no astrocytes; **b**–**d** respectively one, two, and three astrocytes that interact with single synapses (low-order regulation); **e**, **f** one astrocyte that interacts with multiple synapses (higher-order regulation). We show the same data as in Fig. 2 restricted to the monotonic positive regime, for each interaction scheme. Here we show $\langle\nu^{out}\rangle$ as a function of the stimulus frequency. Each data-point corresponds to a value of α, as indicated by the color-code (see color-bar). The median firing rate for each stimulus frequency is shown by the black dashed line. Inset text in each panel reports the average dynamic range Δ (dB) (see Section 6 of Supplementary Information) and the length of the monotonic positive response interval $\Delta\alpha_r$. Parameter values were $\varepsilon = 0.01$, $\mathrm{Ca_{th}} = 0.04$, $\beta = 0.05$ and $D_{\mathrm{Ca}} = 0$. Other simulation parameter values are listed in Table 1.

For $\varepsilon = U_{SE}$, gliotransmission has no net effect, a case known as occlusion[36].

In Fig. 5, we compare decreased and increased facilitation in both lower-order (top panels) and higher-order (bottom panels) interaction schemes. We find, first, that the extension of the SAA regime in α (dark-red area) is not strongly affected by the type of astrocytic modulation (i.e., decreased versus increased facilitation), but rather by the presence of astrocytes and the interaction scheme. Specifically, the extent of the SAA regime in parameter space shrinks under higher-order astrocytic regulation (bottom panels), regardless of whether gliotransmission decreases or increases facilitation. Thus, the mere competition for activating presynaptic neuroreceptors (quantified by γ_{as} and γ_{pre}) is sufficient to prevent (or significantly reduce) vulnerability to runaway activity in the neuronal circuit.

Second, we find that facilitating astrocytic modulation leads to higher firing frequencies in the read-out neuron, as expected. Consequently, the region of excessive suppression that renders the read-out neuron unresponsive (white regions on the right-hand side of the diagrams) is significantly reduced. To illustrate this point, in Supplementary Fig. H we compare different values of $\varepsilon < U_{SE}$ and show that, while the qualitative behavior remains the same, the region of non-responsive read-out neuron progressively shrinks as $\varepsilon \longrightarrow U_{SE}$. Accordingly, the choice of $\varepsilon = 0.01$ is arbitrary, as any circuit with $\varepsilon < U_{SE}$ but sufficiently close to U_{SE} exhibits qualitatively similar behavior across the different modulatory schemes, where the only difference is the size of the unresponsive region for the higher-order modulatory scheme that includes the first (external) synapse.

Exploring different values of ε further reveals that achieving a sensitive read-out response across a large range of stimulus frequencies is possible even when the first synapse is modulated, but would require more precise tuning. Since ε is an effective parameter rather than a directly measurable biophysical quantity, any fine-tuning would be ad hoc and undesirable. Therefore, it is preferable to focus on regimes in which circuit behavior is robust over broad parameter ranges, rather than relying on narrowly tuned values.

Astrocyte diffusive network

The astrocyte model also allows us to introduce coupling between astrocytes through a diffusive term representing gap-junction interactions. Generally, this can have non-trivial effects on the emergent dynamics of the system, particularly in large systems. Here, and in line with the rest of the study, we consider in particular the question of whether astrocyte-coupling in low-

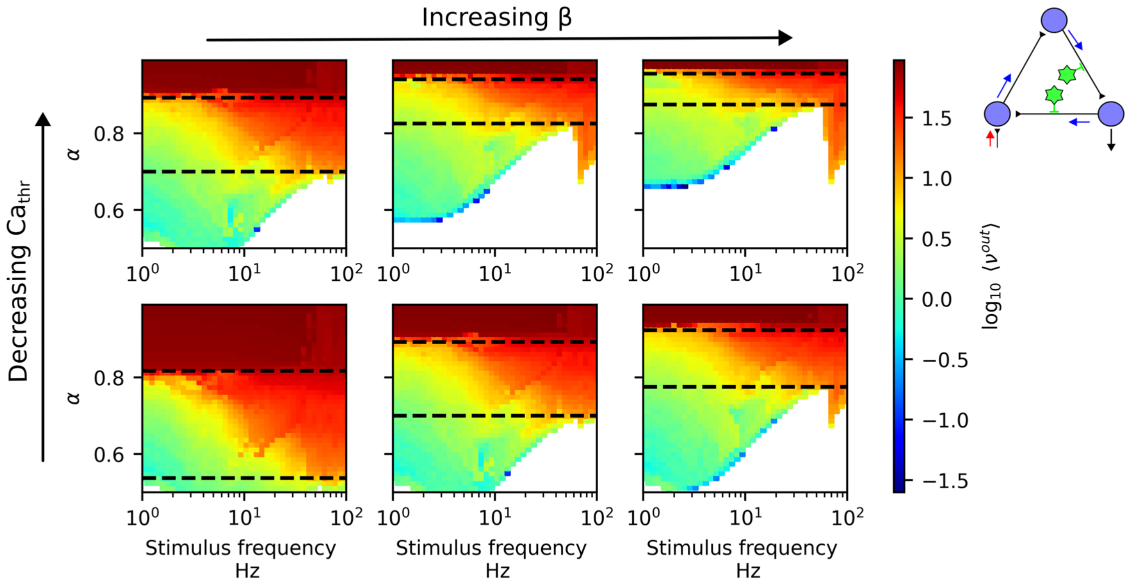

Fig. 4 | Low-order modulation cannot replicate higher-order effect. We show the firing frequency of the read-out neuron $\langle \nu^{out} \rangle$ under a low-order interaction scheme as shown by the graph top-right inset (two uncoupled astrocytes regulating the internal synapses synapses $2 \to 3$ and $3 \to 1$ respectively) for different values of the parameters β and Ca_{th} controlling the astrocytes' calcium dynamics (equation (17)). In particular, $\beta = \{0.05, 0.1, 0.15\}$ for left to right, and $Ca_{th} = \{0.04, 0.02\}$ (bottom to top). The region between the black dashed lines indicates the range of α values for which the read-out neuron responds to the stimulus (below SSA transition) across the explored range of stimulation frequencies. The reference values (considered in the previous figures) are $\beta = 0.05$ and $Ca_{th} = 0.04$. Gliotransmission is facilitated both by increasing β and/or decreasing Ca_{th}. Here $\varepsilon = 0.01$ and all other parameters are as listed in Table 1.

order schemes, in which each astrocyte modulates only one synapse, can replicate the findings of higher-order schemes, in which an astrocyte modulates several synapses concurrently. In Fig. 6, we consider the low-order scheme of three astrocytes, each modulating a synapse, corresponding to panel (c) of Figs. 1 and 2. In this case, we couple the three astrocytes via pair-wise gap-junctions (according to equation (17)), as indicated in the figure, and consider increasing diffusion coefficients D_{Ca}. Our findings show that calcium diffusion across astrocytes has a small effect that does not significantly alter the phase diagram of the system, and, in particular, is not able to recover the effects of higher-order modulation. When a single astrocyte modulates several synapses, it receives and integrates inputs from all of them, resulting in enhanced gliotransmission. Therefore, it is an aggregative, rather than diffusive, effect, and it cannot be replicated via diffusive coupling. In Supplementary Fig. I, we show how diffusive coupling through gap-junctions reduces calcium concentration variability between astrocytes but with minimal effect on plasticity modulation.

However, we emphasize that the limited impact of gap-junction-mediated calcium diffusion observed here is likely specific to the small, local circuits considered in this study. At larger spatial scales, gap-junction coupling and other diffusive mechanisms are expected to play a more prominent role, as they introduce signal propagation delays and spatial gradients that can support richer forms of collective dynamics not observable at very small scales. In extended astrocytic networks, such delays and spatial interactions could enable coordinated or wave-like activity that qualitatively alters neuronal modulation[57,66,67], in contrast to the predominantly aggregative effects observed in the minimal circuits studied here.

Astrocyte modulation in larger networks

In the main part of this study we have considered a simple system made by a three-neuron recurrent circuit. To gain a glimpse on the effects of higher-order astrocyte modulation on larger systems, we consider here larger networks in the form of N-node cycles under different modulation schemes. Firstly, in Fig. 7 we consider N = 5 and four interaction schemes: (a) no astrocyte regulation; (b) a single astrocyte modulating all internal synapses; and (c, d) a single astrocyte modulating two synapses, one of which is the recurrence synapse connecting the read-in and read-out neurons. We find that, in the absence of astrocyte regulation (panel a), the system now enters the SSA regime for almost any value of the control parameters. This behavior is due to the interplay between the intrinsic time-scales of STP recurrent pulses that travel through the cycle, which cause the system to enter a resonance state insensitive to the details of the external stimulus. For more details of this phenomenon see Supplementary Fig. J.

A responsive operating regime becomes possible only in larger cycles through astrocyte modulation. To analyze the effect of particular astrocyte interaction schemes, we focus on those not regulating the input synapse, as we discussed above. We find that, when the astrocyte modulates all synapses but this one (n = 4 interacting synapses, panel b), excessive modulation arises as the frequency of the stimulus increases (white region in the right-hand side of the diagrams) that is akin to the behavior in panel (f) of Fig. 2 and that ultimately disrupts the network's responsiveness to external stimuli. In this case, excessive astrocyte "inhibition" (through decreased facilitation) arises from its interaction with several synapses, which rapidly rises $[Ca^{2+}]$ in the astrocyte. Given that the astrocyte regulates all synapses concurrently, it stops activity propagation before it reaches the read-out neuron.

The case of intermediate higher-order modulation, i.e., when the astrocyte modulates two synapses (n = 2), one of which is the recurrence one and the other is an internal synapse, achieves optimal response of the read-out neuron (panels e and f). In this manner, the astrocyte effectively integrates information about overall network activity through the later synapse, while directly modulating recurrent excitability with the former. This interplay enables an optimal operational regime that balances propagation and stability. Notably, when the astrocyte only modulates internal synapses (panels c for n = 3 and d for n = 2), excessive modulation still occurs.

To gain more insight into the effects of regulating the recurrence synapses, extended the analysis of astrocyte modulation of two synapses to a larger cycle consisting of 20 excitatory neurons in Fig. 8. We considered in particular three schemes: two internal synapses (panel a), the recurrence synapse and the one before it (panel b), and the recurrence synapse and an internal synapse close to the read-in neuron (panel c). We consider in this case the average firing rates of the read-in neuron $\langle \nu^{in} \rangle$ (second row of panels), an intermediate neuron, indicated by the orange arrow in the graph

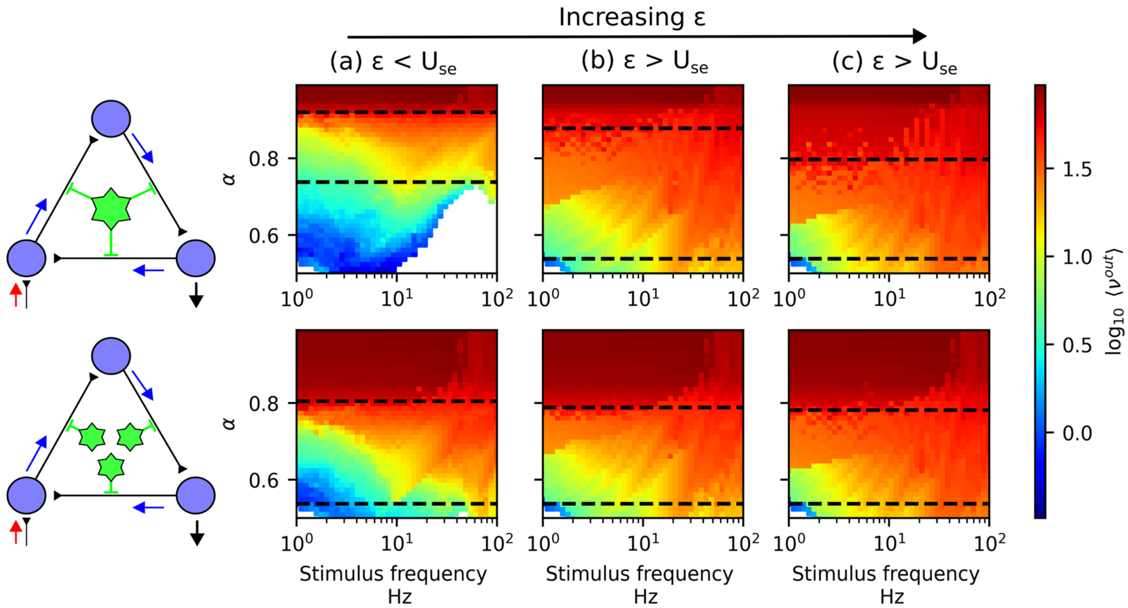

Fig. 5 | From negative to positive astrocyte modulation (effect of ε). Effect of the modulation type (controlled by ε) for a higher-order (top panels) and the corresponding low-order (bottom panels) schemes, as indicated by the graph diagrams. **a** corresponds to decreased facilitation, $\varepsilon = 0.05 < U_{SE}$, whereas **b**, **c** correspond to increased facilitation ($\varepsilon > U_{SE}$), respectively with $\varepsilon = 0.15$ and 0.2. The region between the black dashed lines indicates the range of α values for which the read-out neuron responds to the stimulus (below SSA transition) across the explored range of stimulation frequencies. The reference value (used for the previous figures) is $\varepsilon = 0.01$. Other parameter values were $\mathrm{Ca_{th}} = 0.04$, $\beta = 0.05$ and $D_{\mathrm{Ca}} = 0$. Additional simulation parameter values are provided in Table 1.

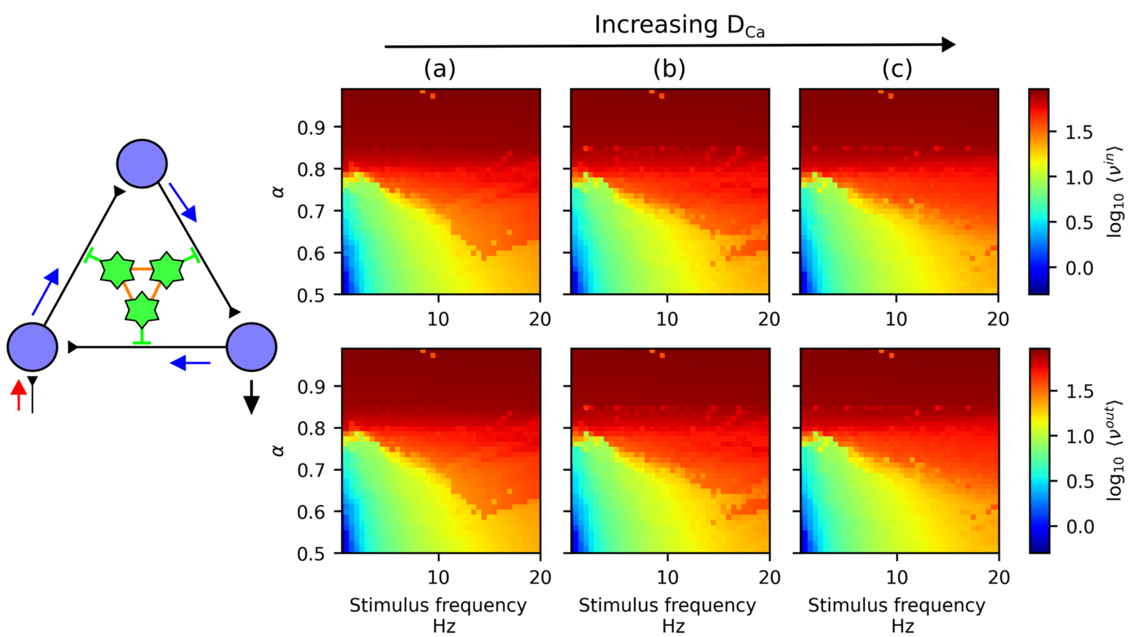

Fig. 6 | Astrocyte diffusion-network does not recover higher-order modulation schemes. Here we consider the low-order modulation scheme with three astrocytes, such that each astrocyte modulates one synapse (corresponding to (**c**) in Figs. 1 and 2), as shown by the graph diagram. In this case the astrocytes are coupled via gap-junctions (equation (17)) as represented by red lines in the graph cartoon. Each column corresponds to a different diffusion coefficient D_{Ca}, namely $D_{\mathrm{Ca}} = 10^{-3}\ \mathrm{ms}^{-1}$ (**a**), $10^{-2}\ \mathrm{ms}^{-1}$ (**b**), and $10^{-1}\ \mathrm{ms}^{-1}$ (**c**). For each panel, we show via a colormap the average firing rate of (top row) the read-in neuron $\langle \nu^{in} \rangle$ and (bottom row) the read-out neuron $\langle \nu^{out} \rangle$. Parameter values were $\varepsilon = 0.01$, $Ca_{th} = 0.04$ and $\beta = 0.05$ in ms^{-1}. Other simulation parameter values are listed in Table 1.

panels, $\langle \nu^{inter} \rangle$ (third row of panels), and the read-out neuron $\langle \nu^{out} \rangle$ (fourth row of panels). As expected from our previous results, the read-in neuron exhibits a safe response to external stimuli of any frequency and across a broad range of α values, due to higher-order astrocyte modulation (see top row of Fig. 8). As before, we observe that increased responsiveness of the read-out neuron occurs when the recurrence synapse is modulated by the astrocyte (panels b and c), regardless of which other synapse is modulated. In contrast, when the astrocyte modulates only two of the first synapses in the circuit (panel a), signal propagation is suppressed for a large region of the parameter space and stimulus frequency; as a result, the read-out neuron is unresponsive (white region in the right-hand side of the read-out neuron diagram).

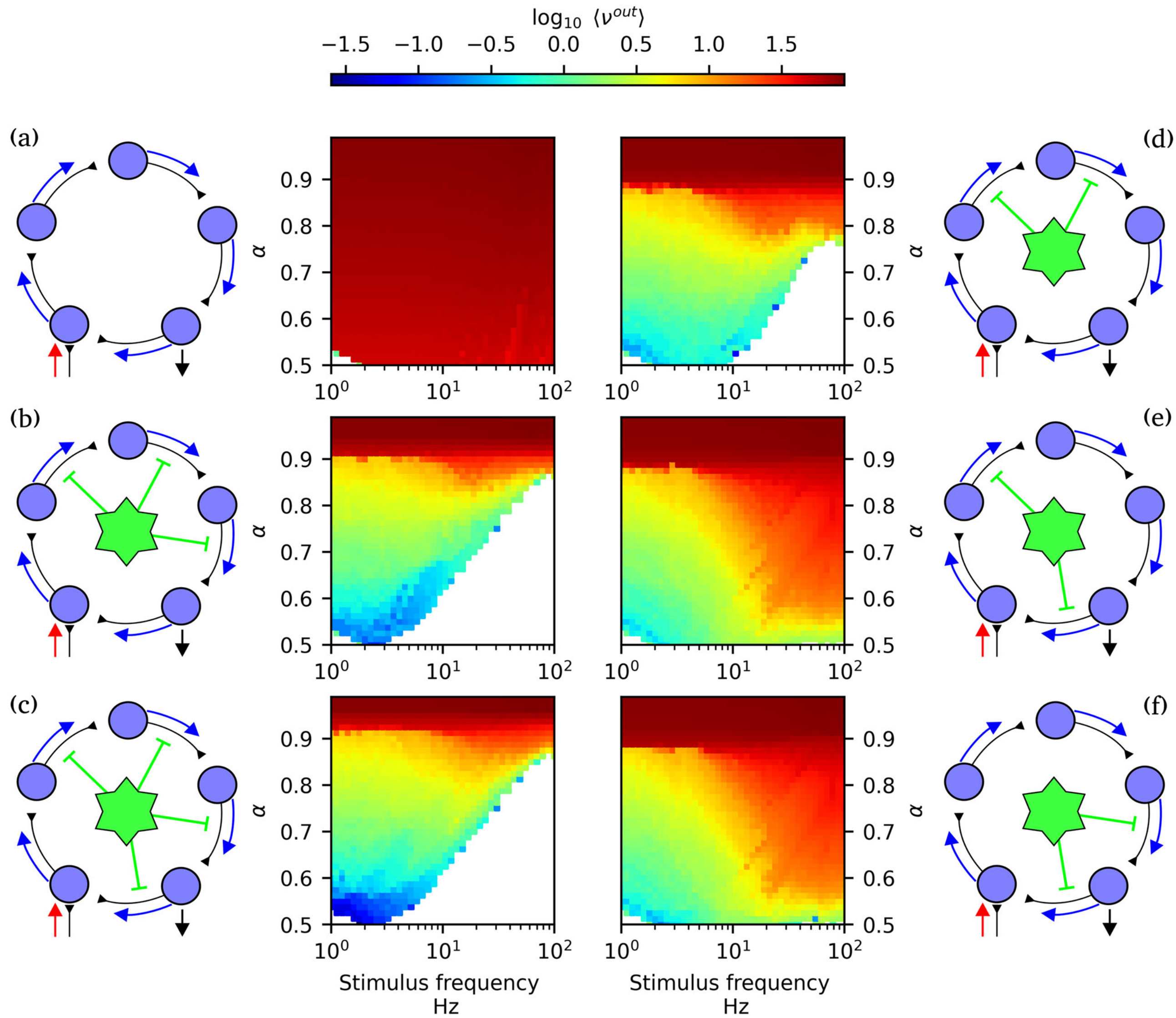

Fig. 7 | Astrocyte modulation in a larger cycle of five neurons. Colormaps indicate the average firing rate of read-out neuron $\langle\nu^{out}\rangle$. Each panel corresponds to a different interaction scheme, as indicated by the graph diagrams: **a** no astrocytes; a single astrocyte modulating (**b**) three internal synapses, (**c**) all four internal synapses; just two internal synapses, where in (**d**) astrocite does not modulate the recurrence synapse and in (**e**, **f**) it modulates the recurrence synapse from the read-out to the read-in neuron, and a second one. Simulation with $\varepsilon = 0.01$, $Ca_{th} = 0.04$, $\beta = 0.05$ and $D_{Ca} = 0$. Other simulation parameter values are listed in Table 1.

To investigate the robustness of these results over the network, we consider now the activity of the internal synapse. Interestingly, when the recurrence synapse is modulated the activity of the internal synapses does depend on the location of the second modulated synapse. When the astrocyte modulates the last two synapses in the cycle (panel b of Fig. 8), activity in the early part of the circuit is weakly modulated, and the intermediate neuron behaves similarly to the read-in neuron. In contrast, if the second synapse modulated by the astrocyte is located early-on in the cycle (panel c of Fig. 8), the behavior of the intermediate neuron resembles now that of the read-out neuron. This result suggests that the astrocyte's spatial scope of integration—whether it spans the beginning and end of the circuit, i.e., accessing non-local information about the network state, or only the end, i.e., accessing only local information—influences activity propagation through the cycle, and the responsiveness of the internal neurons to the external stimuli.

Discussion and conclusions

In recent years, there has been a growing recognition of the active role astrocytes play in shaping neuronal network dynamics[68]. Far from being passive support cells, astrocytes are now understood to participate in a wide range of processes, from neural circuit development[69] to regulation of neural circuit activity[37]. In particular, their involvement in synaptic plasticity—through the release of gliotransmitters and the modulation of pre- and postsynaptic activity—has attracted increasing attention[70,71]. For instance, glutamate clearance, primarily carried out by astrocytes[72], is essential for normal brain function and to prevent excitoxicity which has been hypothesized to play a role in neurodegenerative diseases including amyotrophic lateral sclerosis, Alzheimer's disease and Huntington's disease[73]. Furthermore, gliotransmission can activate extrasynaptic inhibitory receptors through astrocyte GABA release, a mechanism that, when dysregulated, has been linked to Alzheimer's disease[49]. These and other effects highlight astrocyte-neuron interactions as key regulators of both physiological and pathological behaviors in neuronal media. This accumulating experimental evidence has led to a surge in computational[55] and experimental models[74] aiming to capture the bidirectional interactions between astrocytes and neurons.

Despite this momentum, relatively little effort has been made to analyze these neuron-astrocyte interactions from the perspective of *higher-*

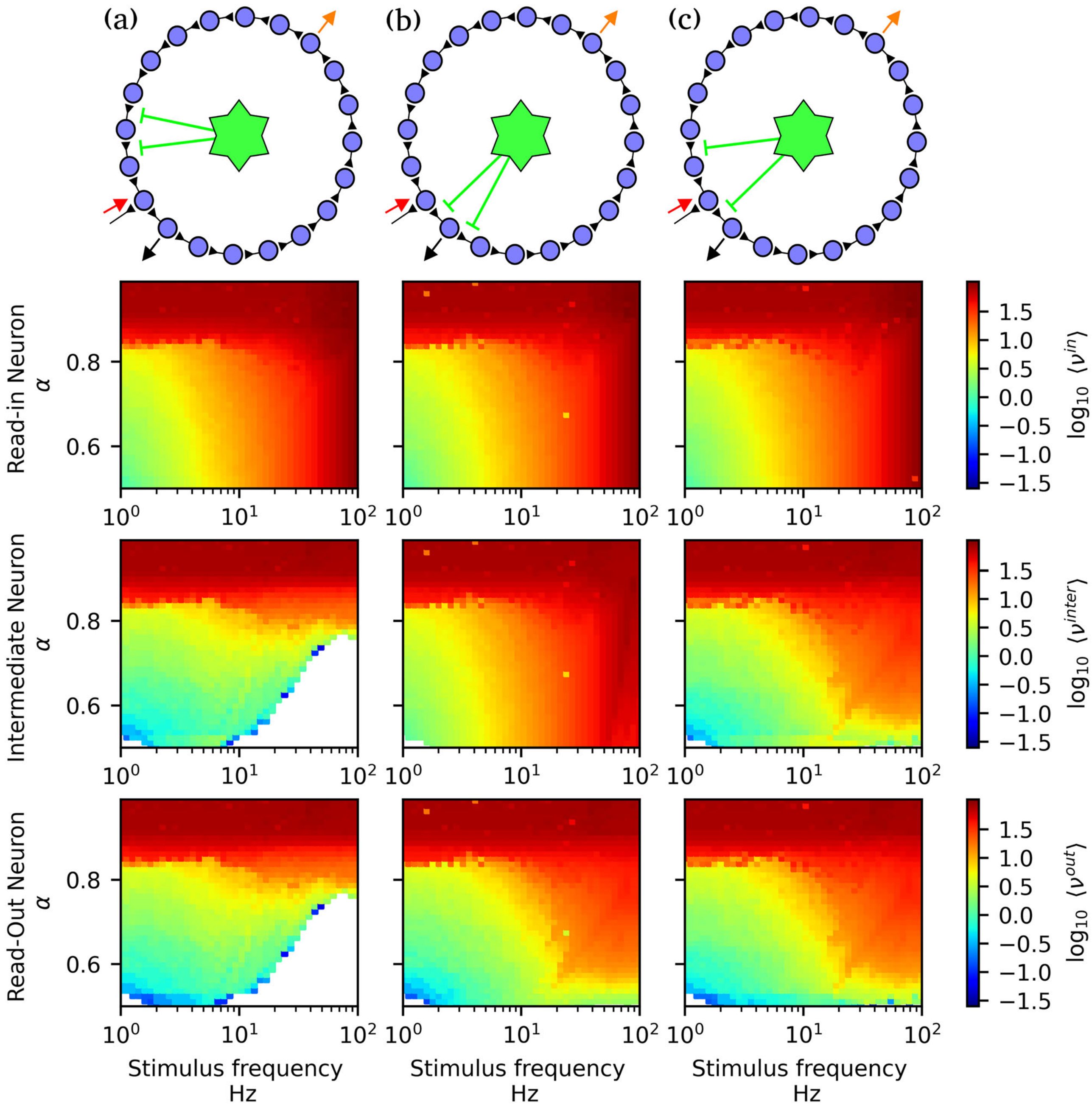

Fig. 8 | Interaction architecture shapes internal response in larger systems. Colormaps indicate the average firing rate of the read-in $\langle \nu^{\mathrm{in}} \rangle$ (top), intermediate $\langle \nu^{\mathrm{inter}} \rangle$ (middle), and read-out $\langle \nu^{\mathrm{out}} \rangle$ (bottom) neurons. Each column corresponds to a different astrocyte interaction scheme, illustrated by the graph diagrams at the top. Red, orange, and black arrows indicate the read-in, intermediate, and read-out neurons, respectively. In all cases, the astrocyte modulates two synapses: **a** not including the recurrence synapse; **b** the recurrence synapse and the previous synapse; and **c** the recurrence synapse and the first internal synapse. Simulations were performed with $\varepsilon = 0.01$, $\mathrm{Ca_{th}} = 0.04$, $\beta = 0.05$, and $D_{\mathrm{Ca}} = 0$. Other simulation parameters are listed in Table 1.

order interactions—that is, interactions that cannot be reduced to the sum of pairwise components[75]. Higher-order interactions are increasingly recognized as a key phenomenon underlying the structure and dynamics of complex systems such as the brain[1,75]. From a modeling perspective, higher-order interactions are known to lead to explosive phase transitions and to change the location of the transition points in the parameter space[40,76], with important consequences for applications e.g., in clinical neuroscience[5,14,77]. In biological networks, higher-order interactions can manifest naturally when a single element such as an astrocyte modulates multiple connections simultaneously, effectively coupling synapses. This is particularly relevant in the context of astrocytes, whose broad spatial reach and integrative capacity allow them to coordinate the activity of multiple synapses within their domain[78]. Interestingly, many models in the literature already implement mechanisms that can be interpreted as higher-order, albeit implicitly. For example, when an astrocyte signal simultaneously alters the release probability or efficacy of two different synapses[50], this constitutes an edge-edge interaction—a classical case of higher-order modulation in a higher-order network. However, such mechanisms are rarely framed or analyzed explicitly through the lens of higher-order network-theory. By making this connection explicit, our study offers a new interpretive lens to understand the computational role of astrocytes. We show that, when a single astrocyte regulates multiple synapses within a recurrent circuit, it introduces non-trivial dependencies that shape local network behavior—dependencies that cannot be decomposed into pairwise synaptic dynamics alone. This reframing helps to bridge cellular-level mechanisms and meso-scale network dynamics, suggesting that glial modulation operates as a higher-order control structure within the brain.

Simulations in a minimal recurrent circuit demonstrate that astrocyte-mediated plasticity can effectively prevent the emergence of SSA—a pathological regime of runaway excitation and loss of responsiveness[79,80]. Crucially, this homeostatic effect only emerges when the astrocyte modulates multiple synapses simultaneously—i.e., when higher-order interactions are present. In this regime, the astrocyte balances excitability and responsiveness by dynamically regulating synaptic efficacy across the circuit. We have compared low- versus higher-order modulating schemes, demonstrating that, when a single astrocyte modulates several synapses concurrently the parameter interval for monotonic positive response is maximized. This effect cannot be reproduced by low-order schemes, even when adjusting model parameters or including diffusive coupling between astrocytes.

Furthermore, we have shown that this regime is prominent when the astrocyte modulates synapses that are internal to the circuit—particularly the recurrent synapse projecting from the read-out neuron back to the read-in neuron. This phenomenon is robust across larger cyclic networks with $N = 5$ and $N = 20$ neurons and across different higher-order coupling schemes. Although the network sizes studied here remain small compared to typical biological neuronal systems, our findings are compatible with the idea that astrocytes can coordinate the modulation of multiple synapses within local circuits in a structured manner. However, we leave open the question of how such mechanisms scale to system-level organization in larger networks.

Our results also align with observations that tripartite synapses are more abundant in brain regions where recurrent connectivity is functionally important—such as the hippocampus and cerebellum[59–62]. In these regions, it is thought that avoiding runaway excitatory activity and maintaining circuit responsiveness typically requires mechanisms such as inhibitory feedback[79]. However, based on our findings, alternative or complementary mechanisms involving tripartite synapses and astrocytic modulation may also play an important role.

Our computational model formalizes this perspective by extending classical STP frameworks[28] to include an astrocyte-mediated facilitation mechanism. The resulting tripartite plasticity model generalizes and unifies previous works: when astrocyte modulation is disabled, it reduces to the standard STP model; when pre-synaptic facilitation is removed, it recovers the astrocyte-driven dynamics proposed by Brunel and De Pittà[50]. This makes the model a flexible platform for studying different regimes of short-time synaptic plasticity, ranging from purely neuronal to glia-integrated mechanisms.

From the higher-order perspective, our model takes advantage of the recently proposed framework of triadic percolation[43,81], that studies the dynamic changes in connectivity induced by triadic interactions—those in which a node regulates the status of the link between two other nodes. Recently, this framework was applied to the study of axo-axonic interactions in a toy-model neural medium of excitatory neurons[13], demonstrating that triadic interactions not only lead to dynamic connectivity, but also to complex spatio-temporal patterns of neuronal activity. Here we took inspiration from these studies to propose the modeling of neuron-astrocyte networks explicitly as a higher-order network in which astrocytes directly control the status to node-to-node interactions—the synapses—and are in turn controlled by neuronal and synaptic activity.

Moreover, while the high-order interaction perspective offers a compelling conceptual framework, further work is needed to systematically quantify and classify such interactions in biologically realistic networks. Future studies could expand this framework to larger and more heterogeneous systems, include multiple interacting astrocytes, and explore how astrocyte modulation interfaces with long-term plasticity mechanisms. Additionally, formal tools from higher-order network theory and topological data analysis could be employed to characterize emergent dynamics and identify signatures of glial modulation in experimental data.

This work introduces a computational framework that formalizes astrocyte-mediated modulation of synaptic dynamics as a tripartite plasticity mechanism. By extending classical STP models to include astrocyte-dependent facilitation, our approach captures how a single astrocyte can simultaneously influence multiple synapses, thereby introducing higher-order dependencies into network dynamics. This provides a biologically grounded implementation of higher-order interactions in neuronal-astrocytes systems.

Despite these contributions, our model remains a limited simplification of biological reality. We describe a minimal circuit of excitatory neurons, which does not incorporate inhibitory dynamics, astrocyte heterogeneity, or spatially distributed networks. While focusing on local-scale gliotransmission provides a tractable and mechanistically transparent entry point to study higher-order astrocyte-synapse interactions, we acknowledge that astrocytes can influence neuronal activity through many additional mechanisms, operating at both low- and high-order levels. For example, astrocyte could regulate synaptic efficacy through gliotransmission, but also through neurotransmitter uptake, being able to induce global synchronous bursting[82]. In the context of our model, this means the astrocyte could effectively (in both low- and high- order schemes) manipulate α (our model control parameter), making it a dynamic homeostatic variable instead of a fixed parameter.

Astrocytes can also act directly on postsynaptic neurons by promoting slow inward currents through extrasynaptic receptors[48,83], as well as by inducing inhibitory postsynaptic currents via GABA release[84]. Incorporating these effects would require extending the model to include an explicit astrocyte-neuron current, corresponding to a low-order interaction term in the membrane potential equation (equation (1)). Moreover, astrocytic regulation of extracellular potassium levels can modulate neuronal after-hyperpolarization, a mechanism shown, both theoretically and experimentally, to play a key role in shaping inter-burst interval statistics[39,85], and to depend critically on astrocytic gap-junction coupling at larger spatial scales. Within our framework, these mechanisms would translate into astrocyte-induced modulation of the refractory time constant t_a and motivate extensions toward coupled neuron-astrocyte networks beyond the local synaptic scale explored in our current work.

Overall, this work contributes to a growing body of evidence that astrocytes are key players in neural computation—not only at the synaptic level but also in the emergent coordination of network activity through structured interactions[43,86]. However, our model explores only a restricted region of the broader landscape of possible astrocyte-mediated (low- and high-order) modulation mechanisms. Extending the framework to incorporate additional modulation pathways and larger spatial scales therefore represents an important direction for future work.

Methods

Higher-order neuron-synapse-astrocyte model

Here we propose a simple ASN-STP that captures the potential competition between gliotransmission, mediated by astrocytes, and pre-synaptic mechanisms, mediated by the pre-synaptic neuron, as short-term facilitation. Our model includes the description of neuronal, synaptic, and astrocyte dynamics, as well as the interactions among them. Neurons constitute the nodes of the higher-order network, and thus we refer to their dynamics as dimension-0. Synapses connect neurons in a directed manner, with the pre-synaptic neuron projecting onto the post-synaptic neuron. Synaptic dynamics are represented as 1-dimensional topological signals taking place on the edges of the higher-order network. Finally, astrocytes modulate synaptic dynamics in an aggregative, non-linear manner. In actual brains, one astrocyte modulates several synapses at the same time, whereas each synapse can have at most one adjacent astrocyte, as astrocyte will usually have non-overlapping territories[87]. Given that astrocytes couple the 1-dimensional edge signals, they are represented as 2-dimensional topological signals. An exemplary minimal circuit composed by three neurons arranged in a loop, with three directed excitatory synapses, and one astrocyte modulating the three synapses, is shown in Fig. 9.

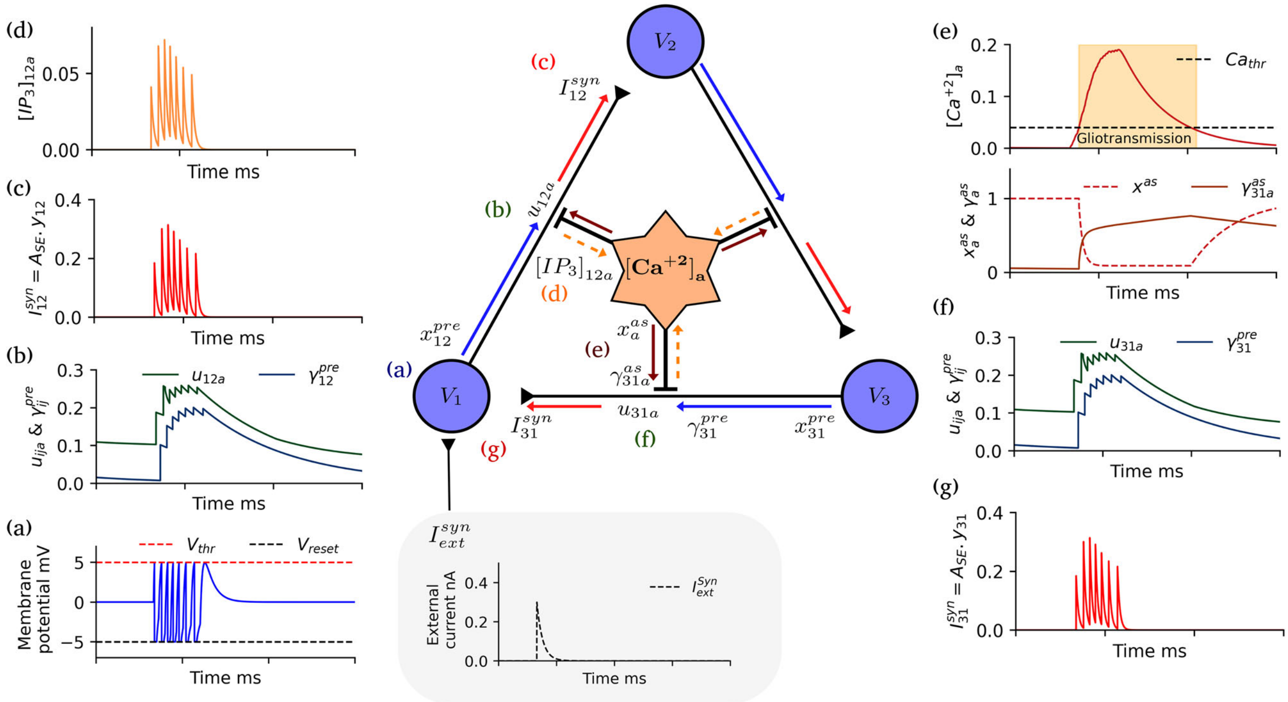

Fig. 9 | Model scheme in a minimal recurrent excitatory circuit. We consider a minimal recurrent network consisting of three excitatory neurons arranged in a ring, with states V_i. An external input I_{ext}^{syn} arrives at neuron 1 inducing spiking activity, as shown in (**a**), which propagates through the circuit. **b** show the synaptic variables u_{12}^a (neurotransmitter release probability) and $y_{12,\mathrm{pre}}^a$ (pre-synaptic facilitation variable), which increase following pre-synaptic activity. **c** shows the amplitude of the synaptic currents at synapse 1 → 2, namely $A_{SE}y_{12}$ (equation (3)), which is affected both by the dynamics of neuron 1 and the astrocyte a. **d** shows the IP_3 dynamics in the astrocyte process at synapse ij, displaying the increase in $[IP_3]_{12}^a$ following pre-synaptic spikes. **e** shows the dynamics of the astrocyte variables: the calcium concentration $[Ca^{+2}]^a$ (top panel), and the astrocyte depletion and facilitation variables x_{astro}^a and $y_{ij,\mathrm{astro}}^a$. High-order interaction (HOI): By astrocyte mediation, the activity on synapse 1 → 2 affects facilitation in synapse 3 → 1 (**f**), which change the recurrence current in Neuron 1 (**g**).

Neuronal, synaptic, and astrocyte dynamics used in the present work are defined in detail in the following subsections. Here we describe them in a summary manner. The reference values of all parameters are detailed in Table 1. Neurons are defined as leaky integrate-and-fire (LIF) units whose state is given by a voltage variable $V_i(t)$ (equation (1)), such that the neuron is said to fire or release an action potential or spike when the voltage reaches a threshold. Afterwards, $V_i(t)$ is reset to a hyperpolarized (negative) value. Synapses are characterized by the fraction of active neurotransmitters y_{ij} (equation (4)). Each time the pre-synaptic neuron fires, y_{ij} increases depending on three factors: the fraction α of neurotransmitters that remain at the synapse, the probability of neurotransmitter release u_{ij}, and the fraction of available neurotransmitters x_{ij}. The parameter α accounts for the competition between the astrocyte and the synapse for released neurotransmitters. A fraction $1-\alpha$ of released neurotransmitter is recruited by the astrocyte (equation (15)), leaving only a fraction α available for synaptic transmision. Thus α is a main control parameter of the dynamics. The synaptic variables u_{ij} (equation (11)) and x_{ij} (equation (12)) account respectively for short-term facilitation and depression mechanisms.

STF is affected both by presynaptic mechanisms driven by the pre-synaptic neuron and gliotransmission driven by the astrocyte. Here, we consider both effects, as well as the competition between them. The neuron-driven plasticity dynamics is inspired by previous models of neuron-driven STP[25,28]. Regarding astrocytes, they affect STF through gliotransmission, the process by which they recruit and release gliotransmitters at each adjacent synapse[50]. Astrocyte dynamics is usually described by variations of its cytosolic calcium concentration, $[Ca^{2+}]^a$. $[Ca^{2+}]^a$ increases following each pre-synaptic neuron spike at any synapse regulated by the astrocyte, from which the astrocyte recruits a fraction of the released neurotransmitters proportional to $1-\alpha$. When a $[Ca^{2+}]^a$ threshold is reached, we assume the astrocyte to become active and release gliotransmitters into its adjacent synapses ij, modulating their pre-synaptic release probability u_{ij} (equation (11)). Contrary to neuron-driven STF, gliotransmission can lead to both an increase and a decrease of the probability of neurotransmitter release depending on a control parameter (namely ε in equation (11))[53]. Moreover, recruitment of neurotransmitters by the astrocyte competes with neuron-driven STF, naturally leading to competition effects that cannot be recovered by the independent original models, as we go on to show. The competition is affected by the relative temporal scales of each mechanisms. Pre-synaptic STF occurs on scales of hundreds of milliseconds to a few seconds, whereas gliotransmission is slower and takes place in the time-scale of seconds to minutes. This fact, and its consequences, is directly captured with our proposed model. Finally, we also consider short-term depression in the model, reflecting the depletion of neurotransmitters after repeated resources release, and which occurs both at the synapse and astrocyte levels (eqs. (12)).

Dimension zero dynamics: the neurons

The neuronal population is modeled using the LIF model. If the membrane potential of the neuron is below a threshold V_{th}, the dynamics follow:

$$\tau_V \frac{dV_j}{dt} = -V_j + RI_j^{stim}(t) + RI_j^{syn}(t), \tag{1}$$

where V_j is the membrane potential of neuron j, R the membrane resistance, and τ_V the membrane time constant. When the membrane potential crosses the threshold, i.e., $V_j(t) \geq V_{th}$ at time $t = t_j^{spk}$, a spike is emitted by neuron j, and the membrane potential is reset to a hyperpolarized value, $V(t_j^{spk}) = V_{reset}$ mV. Following this, the neuron enters an absolute refractory period t_a, during which the potential is clamped at V_{reset}.

Table 1 | Description, symbols, and values of model parameters

Parameter	Symbol	Value
Membrane time constant	τ_V	20 ms
Membrane resistance	R	200 MΩ
Absolute refractory period	t_a	4 ms
Spike threshold potential	V_{th}	5 mV
Reset potential	V_{reset}	−5 mV
Synaptic deactivation time constant	τ_{in}	4 ms
Absolute synaptic efficacy	$\mathcal{A}_{SE}$	3 nA
Fraction of neurotransmitter in the cleft	α	0.25–0.975
Neurotransmitter recovery time constant	τ_r	100 ms
Gliotransmitter recovery time constant	τ_r^{as}	100 ms
Steady-state gliotransmitter release probability	U_{SE}^{as}	0.1
Steady-state neurotransmitter release probability	U_{SE}	0.1
Astrocyte facilitation modulation parameter	ε	0.01–0.2
IP_3 decay time constant	τ_{IP_3}	6 ms
Calcium decay time constant	τ_{Ca}	100 ms
Astrocyte calcium diffusion coefficient	D_{Ca}	$0-10^{-1}$ ms^{-1}
Calcium increase rate via IP_3	β	0.05−0.15 ms^{-1}
Synaptic facilitation time constant (pre-synaptic)	τ_f^{pre}	200 ms
Synaptic facilitation time constant (astrocyte)	τ_f^{as}	5000 ms
Gliorelease calcium threshold	Ca_{th}	0.02−0.04 μM
External synaptic current amplitude	A_{ext}	0.3 nA

To ensure physiologically plausible synaptic currents, $\mathcal{A}_{SE}$ is tuned such that the peak current from an individual synapse is of order $\mathcal{O}(10^{-1})$ nA. For example, typical AMPA receptor-mediated synaptic currents are around 0.3 nA[98]. Given that both x_{ij} and u_{ij} are on the order of $\mathcal{O}(10^{-1})$, it follows that $\mathcal{A}_{SE}$ should be of order $\mathcal{O}(1)$ nA to yield realistic current magnitudes.

An input current $I_1^{stim}(t)$ arrives at neuron 1 ($I_{j\neq 1}^{stim}(t)=0$), which is modeled as a series of exponentially decaying pulses applied at specific times t_k^{stim}, namely

$$I_1^{stim}(t)=\sum\nolimits_k A_{ext}\exp\left(\frac{t_k^{stim}-t}{\tau_{in}}\right)\Theta(t-t_k^{stim}), \tag{2}$$

where A_{ext} is the stimulus amplitude, τ_{in} the decay time constant, and $\Theta(\cdot)$ the Heaviside step function. The synaptic current received by neuron j is, including STP mechanisms:

$$I_j^{syn}(t)=\sum\nolimits_{i\in nn(j)}\mathcal{A}_{SE}\,y_{ij}(t), \tag{3}$$

where $nn(j)$ denotes the set of pre-synaptic neurons projecting onto neuron j, and $\mathcal{A}_{SE}$ is the maximum synaptic efficacy. The variable $y_{ij}(t)$ indicates the synaptic efficacy: it stands for fraction of active neurotransmitters in synapse ij, i.e., the amount of neurotransmitters that bound to post-synaptic receptors. The dynamics of $y_{ij}(t)$ depend both on node- and astrocyte-driven short-term mechanisms, as we describe in the following subsections.

Dimension one dynamics: the synapses

Lets consider first that each synapse is modulated by one astrocyte. In the following we indicate this in all synaptic variables by the super-index a, indicating the astrocyte that modulates the synapse. The fraction of active neurotransmitter $y_{ij}^a(t)$ accounting for the dynamical state of synapse ij, modulated by astrocyte a, evolves according to (see[28]):

$$\frac{dy_{ij}^a}{dt}=-\frac{y_{ij}^a}{\tau_{in}}+\alpha\, x_{ij}^a(t)\,u_{ij}^a(t)\,\delta(t-t_i^{spk}), \tag{4}$$

Here α is a main control parameter of the model, representing the fraction of neurotransmitters that remain in the synaptic cleft and contribute to the post-synaptic current (equation (3)). A fraction $1-\alpha$ of neurotransmitter escapes the synaptic cleft and activates astrocyte receptors instead of post-synaptic ones. The explicit inclusion of α thus allows us to differentiate between synaptic and astrocyte activation. The synaptic variables $x_{ij}^a(t)$ and $u_{ij}^a(t)$ account respectively for the fraction of available synaptic resources (neurotransmitters) and for the neurotransmitter release probability at synapse ij, and account for STP mechanisms. In the following, we discuss firstly facilitation and secondly depression mechanisms.

Short-term facilitation: node- and astrocyte-driven mechanisms. To model STF when both pre-synaptic and gliotransmission mechanisms are present, we extend the strategy presented in ref. 50. Firstly, we hypothesize that $u_{ij}^a(t)$ depends on the fraction of receptors and channels activated by (a) gliotransmission and (b) pre-synaptic activity, namely $\gamma_{ij,\mathrm{astro}}^a(t)$ and $\gamma_{ij,\mathrm{pre}}^a(t)$. Secondly, in the absence of neuronal of astrocyte inputs, $u_{ij}^a(t)$ should reach a steady-state U_{SE} (where u stands for "utilization" and SE for "synaptic efficacy"), as originally introduced in the Tsodyks-Markram model for STP[25,28]. We further hypothesize that, near its steady state, $u_{ij}^a(t)$ is a smooth function that we can approximate by its first-order power expansion, i.e.,

$$u_{ij}^a(\gamma_{ij,\mathrm{pre}}^a,\gamma_{ij,\mathrm{astro}}^a,t)\approx U_{SE}+\left.\frac{\partial u_{ij}^a}{\partial\gamma_{ij,\mathrm{astro}}^a}\right|_{(0,0)}\gamma_{ij,\mathrm{astro}}^a(t)+\left.\frac{\partial u_{ij}^a}{\partial\gamma_{ij,\mathrm{pre}}^a}\right|_{(0,0)}\gamma_{ij,\mathrm{pre}}^a(t). \tag{5}$$

An important detail is the timescale difference between the two mechanisms: pre-synaptic facilitation typically operates over hundreds of milliseconds to a few seconds, whereas gliotransmission can influence synaptic function over longer timescales (seconds to minutes)[36,50]. This justifies modeling both activated fractions with separate dynamics and distinct time constants.

The fractions of active neurotransmitters are bounded between 0 and 1, as well as the total fraction, i.e., $0\le\gamma_{ij,\mathrm{pre}}^a(t)+\gamma_{ij,\mathrm{astro}}^a(t)\le 1$. Moreover, given that u_{ij}^a is a probability, it is also bounded between 0 and 1, and the derivatives in equation (5) must be defined to maintain these bounds. In particular, for the pre-synaptic mechanism we set

$$\left.\frac{\partial u_{ij}^a}{\partial\gamma_{ij,\mathrm{pre}}^a}\right|_{(0,0)}=1-U_{SE}, \tag{6}$$

which recovers the Tsodyks-Markram dynamics in the absence of gliotransmission. Regarding gliotransmission, we follow the reasoning in ref. 36 and define

$$\left.\frac{\partial u_{ij}^a}{\partial\gamma_{ij,\mathrm{astro}}^a}\right|_{(0,0)}=\varepsilon-U_{SE}, \tag{7}$$

where $0<\varepsilon<1$. This allows us to model the dual nature of gliotransmission, i.e., its capacity to either promote or inhibit neurotransmitter release. In particular, if $\varepsilon > U_{SE}$, gliotransmission increases the release probability; otherwise, it decreases it. We note that these definitions ensure that u^a_{ij} is bounded between 0 and 1 both in the case of no transmission (in which case $\gamma^a_{ij,\mathrm{astro}} = \gamma^a_{ij,\mathrm{pre}} = 0$ and $u^a_{ij} = U_{SE}$), and maximum transmission ($\gamma^{pre}_{ij} + \gamma^a_{ij,\mathrm{astro}} = 0$).

Next, we define the dynamics of the fractions of activated neurotransmitters. Assuming that the onset of receptor activation's effect on synaptic release is much faster than its decay time for both mechanisms, the coupled dynamics are described as:

$$\frac{d\gamma^a_{ij,\mathrm{astro}}}{dt} = -\frac{\gamma^a_{ij,\mathrm{astro}}}{\tau_{f,\mathrm{astro}}} + G^a(t)\left[1 - (\gamma^a_{ij,\mathrm{astro}} + \gamma^a_{ij,\mathrm{pre}})\right]\Theta([\mathrm{Ca}^{2+}]^a(t) - \mathrm{Ca_{th}}), \tag{8}$$

$$\frac{d\gamma^a_{ij,\mathrm{pre}}}{dt} = -\frac{\gamma^a_{ij,\mathrm{pre}}}{\tau_{f,\mathrm{pre}}} + U_{SE}\left[1 - (\gamma^a_{ij,\mathrm{astro}} + \gamma^a_{ij,\mathrm{pre}})\right]\delta(t - t^{sp}_i), \tag{9}$$

where $G^a(t)$ is the gliotransmitter release efficacy, which is proportional to the availability of gliotransmitters at the astrocyte[53], $x^a_{\mathrm{astro}}(t)$, namely

$$G^a(t) = x^a_{\mathrm{astro}}(t)U_{\mathrm{astro}}, \tag{10}$$

where U_{astro} is a constant setting the release probability. Gliotransmitter release is assumed to be a continuous process occurring as long as the astrocyte calcium concentration remains above the threshold $\mathrm{Ca_{th}}$ and resources are available. In contrast, the pre-synaptic facilitation mechanism —driven by voltage-dependent calcium channels—is triggered directly by the arrival of an action potential and is independent on the amount of neurotransmitter concentration in the synaptic cleft. Thus, we use the same efficacy U_{SE}, which ensures that in the absence of gliotransmission, the mechanism recovers the original facilitation model proposed by refs. 25,28.

The release probability is ultimately expressed as

$$u^a_{ij}(\gamma^a_{ij,\mathrm{astro}}, \gamma^a_{ij,\mathrm{pre}}, t) = U_{SE} + (\varepsilon - U_{SE})\gamma^a_{ij,\mathrm{astro}}(t) + (1 - U_{SE})\gamma^a_{ij,\mathrm{pre}}(t), \tag{11}$$

where one needs to solve the coupled dynamics of $\gamma^a_{ij,\mathrm{pre}}(t)$ and $\gamma^a_{ij,\mathrm{astro}}(t)$. Our proposed facilitation model generalizes previous models as we show in the Supplementary Information (Section 1). In particular, we demonstrate how our model reduces to the classical facilitation model proposed in refs. 25,28 in the absence of gliotransmission, and how it becomes equivalent to the model in ref. 50 in the absence of pre-synaptic facilitation.

Resource depletion. Due to resource depletion, neither neurotransmission nor gliotransmission can occur indefinitely under repeated stimulation. To model this, we consider a depression mechanism such that the fraction of available neurotransmitters and gliotransmitters decreases with each release event. The former is implemented via a short-term depression model for the pre-synaptic spine[28], and we introduce a novel analogous mechanism for the latter:

$$\frac{dx^a_{ij}}{dt} = \frac{1 - x^a_{ij}(t)}{\tau_r} - u^a_{ij}(t)\,x^a_{ij}(t)\,\delta(t - t^i_{sp}) \tag{12}$$

$$\frac{dx^a_{\mathrm{astro}}}{dt} = \frac{1 - x^a_{\mathrm{astro}}(t)}{\tau_{r,\mathrm{astro}}} - U_{\mathrm{astro}}\,x^a_{\mathrm{astro}}(t)\,\Theta([\mathrm{Ca}^{2+}]_a - \mathrm{Ca_{th}}), \tag{13}$$

where τ_r and $\tau^{r,\mathrm{astro}}$ are the recovery time constants for synaptic and astrocyte processes, respectively. In this model, we assume these values to be uniform across all processes. The synaptic resource variable $x^a_{ij}(t)$ decreases instantaneously with every pre-synaptic spike t^i_{sp}, and recovers exponentially between spikes with time constant τ_r. Similarly, the astrocyte resource variable $x^a_{\mathrm{astro}}(t)$ depletes proportionally to gliotransmitter release, which occurs when astrocyte calcium concentration exceeds the threshold $\mathrm{Ca_{th}}$, and recovers with time constant $\tau_{r,\mathrm{astro}}$. This formulation ensures that both synaptic and astrocyte release dynamics are constrained by a limited pool of resources.

Dimension two dynamics: the astrocytes

Astrocytes possess two main compartments: the soma and the astrocyte processes, i.e., the terminal structures (leaflets) that directly interact with neuronal synapses and form the tripartite synapse[88]. Each process can host localized calcium domains with faster dynamics compared to the soma[35]. Calcium can diffuse from one process to another and to the soma, where the signals from different processes are integrated, enabling a global calcium response and territorial synaptic modulation[78]. This means that an astrocyte can modulate one synapse based on activity at other synapses within its territory of influence. Here, we consider a simplified scenario where calcium diffuses instantaneously within the cell, reducing the complexity of the model, allowing us to focus on the integration effect of the astrocyte.

We describe the state of astrocyte a by its calcium concentration $[\mathrm{Ca}^{2+}]^a(t)$. Similarly to neuronal dynamics, when $[\mathrm{Ca}^{2+}]^a(t)$ crosses a threshold $\mathrm{Ca_{th}}$ we assume the astrocyte to become active and starts to release gliotransmitters into its neighboring synapses, modulating the presynaptic release probability $u^a_{ij}(t)$, as shown in the previous section (eqs. (9) and (11)). Unlike neuronal dynamics and previous works that model this as a discrete event[50,58] (i.e., a spike), we treat gliotransmitter release as a continuous process. This is supported by experimental and computational evidence indicating that astrocyte calcium signals are slow (lasting from hundreds of milliseconds to several seconds or more) compared to the millisecond-scale action potentials in neurons[56,89–91]. Therefore, we consider an astrocyte to be active as long as $[\mathrm{Ca}^{2+}]^a(t)$ remains above the threshold $\mathrm{Ca_{th}}$ (eqs. (9) and (12)).

We consider in our model three main mechanisms driving $[\mathrm{Ca}^{2+}]^a(t)$ dynamics. Firstly, $[\mathrm{Ca}^{2+}]^a(t)$ increases following each pre-synaptic spike in the synapses ij modulated by astrocyte a. This occurs through the release of glutamate caused by the firing of the pre-synaptic neuron, which in turn leads to the synthesis of IP_3 (Inositol Trisphosphate, a signaling molecule) inside the astrocyte[92]. Following the formulation proposed by ref. 58, which simplifies earlier work[93], we assume that the concentration of IP_3 at the astrocyte processes of astrocyte a that interact which synapse ij, $[IP_3]^a_{ij}$, increases instantaneously whenever the presynaptic neuron i fires an action potential, namely

$$\frac{d[IP_3]^a_{ij}}{dt} = -\frac{[IP_3]^a_{ij}}{\tau_{IP_3}} + W^a_{ij}(t)\left[1 - [IP_3]^a_{ij}\right]\delta(t - t^i_{sp}), \tag{14}$$

where the term

$$W^a_{ij}(t) = (1 - \alpha)x^a_{ij}(t)u^a_{ij}(t) \tag{15}$$

represents the coupling efficacy between the synapse ij and the astrocyte process. Here, $(1 - \alpha)$ denotes the fraction of neurotransmitters that escapes the synaptic cleft and binds to receptors on astrocyte processes (such as glutamate metabotropic receptors). We note that this formulation constrains $[IP_3]$ between 0 and 1, serving both as a normalization and a simple way to incorporate saturation. As discussed above, we assume that IP_3 diffuses instantaneously within the astrocyte, and thus we can write the total IP_3 concentration in astrocyte a, $[IP_3]^a$, as

$$[IP_3]^a = \sum\nolimits_{ij\in nn(a)}[IP_3]^a_{ij}(t), \tag{16}$$

where $nn(a)$ denotes the set of synaptic neighbors, i.e., all synapses adjacent to the astrocyte.

The second mechanism in the astrocyte calcium dynamics is calcium release or clearance away from the soma and astrocyte-synapse processes. Different mechanisms are at play in this processes, including e.g., SERCA pumps that transport calcium into the astrocyte's endoplasmatic reticulum (where it is not readily available for gliotransmission)[56]. Here, we simplify these processes to an effective exponential decay of $[\mathrm{Ca}^{2+}]^a$ over time, following approaches used in previous models for both synaptic and astrocyte calcium dynamics[50,94,95]. Finally, the third mechanism of astrocyte calcium dynamics accounts for direct interaction between astrocytes. Astrocytes are interconnected through gap junctions, forming undirected astrocyte networks that enable calcium diffusion between cells[96]. Considering the three mechanisms together, we arrive at the following dynamics for the calcium concentration at astrocyte a:

$$\frac{d[\mathrm{Ca}^{2+}]^a}{dt} = -\frac{[\mathrm{Ca}^{2+}]^a}{\tau_{\mathrm{Ca}}} + \beta [IP_3]^a(t) + \sum\nolimits_{k \in ann(a)} D_{\mathrm{Ca}} \left([\mathrm{Ca}^{2+}]^k - [\mathrm{Ca}^{2+}]^a \right), \tag{17}$$

where β has units of ms^{-1}, τ_{Ca} is a decay constant that captures effective calcium clearance due to calcium pumps, and D_{Ca} is the diffusion coefficient modeling gap-junction-mediated calcium exchange with astrocyte neighbors $ann(a)$.

The coupled system of differential equations was integrated numerically using a forward Euler method with a fixed time step of 10^{-3}. Simulations were implemented in `Julia`. We verified that further reductions of the time step did not qualitatively affect the results. The source code used in this study is available in a public GitHub repository (see[97]).

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data supporting the findings of this study were generated from numerical simulations of the model described in the "Methods" section. These data can be reproduced using the code provided and are therefore not separately archived. Example outputs are available from the corresponding author upon reasonable request.

Code availability

The code used to perform the simulations and generate the results is publicly available[97].

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Author contributions

A.P.M., G.M. and J.J.T. conceived the study. G.M. and J.J.T. developed the theoretical framework and computational methodology, and designed the numerical experiments. G.M. implemented the model, performed the simulations, conducted the data analysis, visualization and interpretation of the results. G.M., A.P.M. and J.J.T. wrote the original draft of the manuscript. G.M. and J.J.T. contributed to manuscript revision and provided critical feedback. J.J.T. supervised the project, and acquired funding.

Competing interests

The authors declare no competing interests.

Additional information

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Correspondence and requests for materials should be addressed to Gustavo Menesse or Joaquín J. Torres.

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